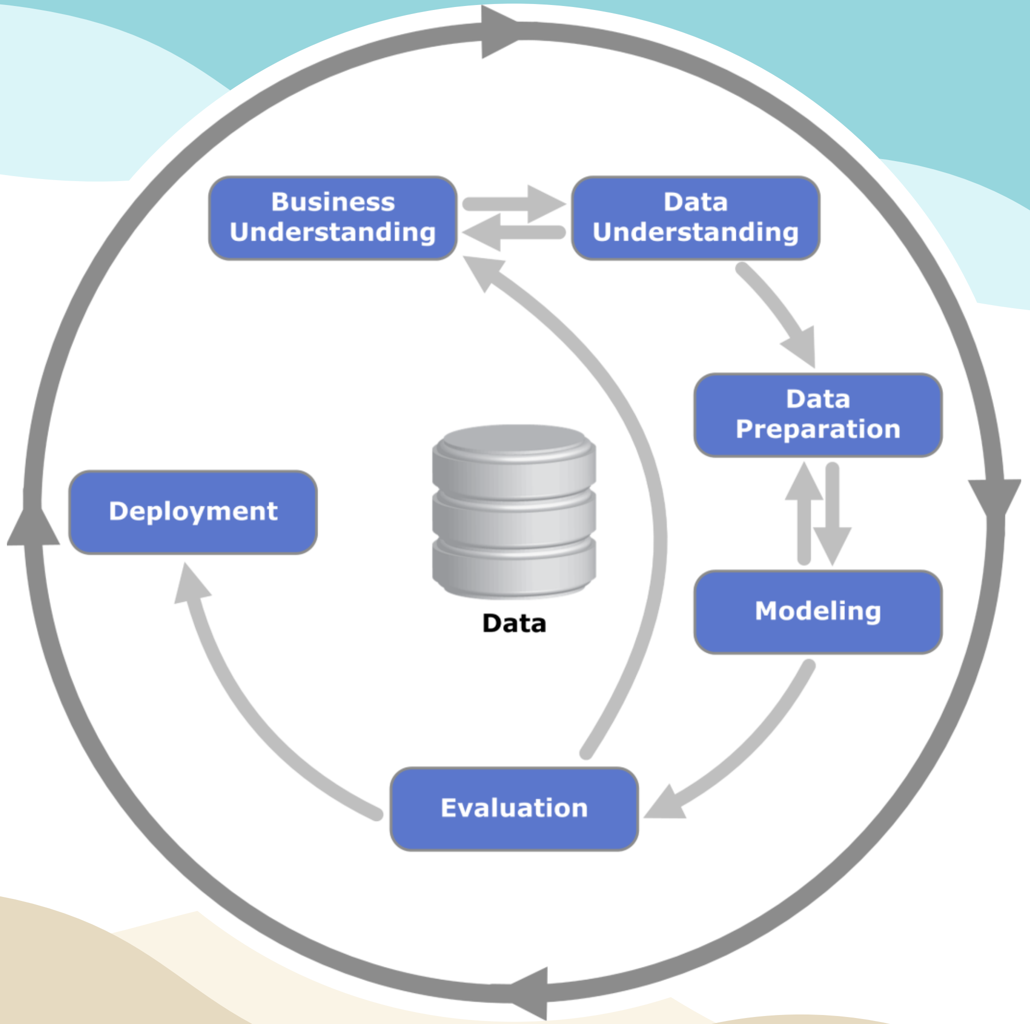
My Hero's Journey

Automated Resistor Value Recognition

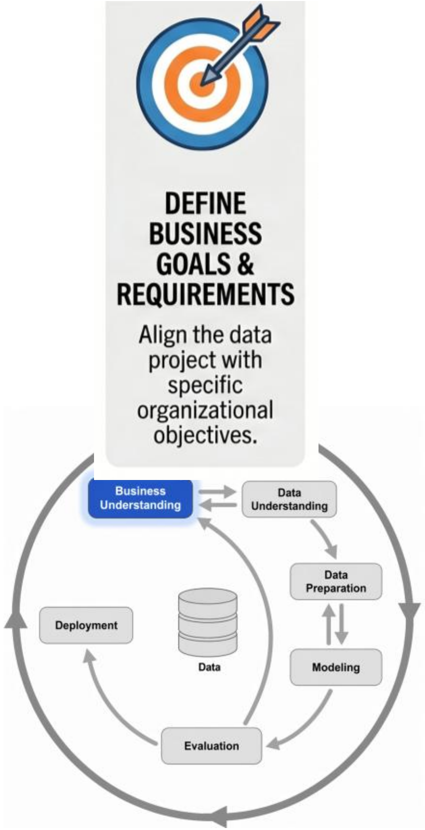
Hassan Alshaikh

RCEL 506

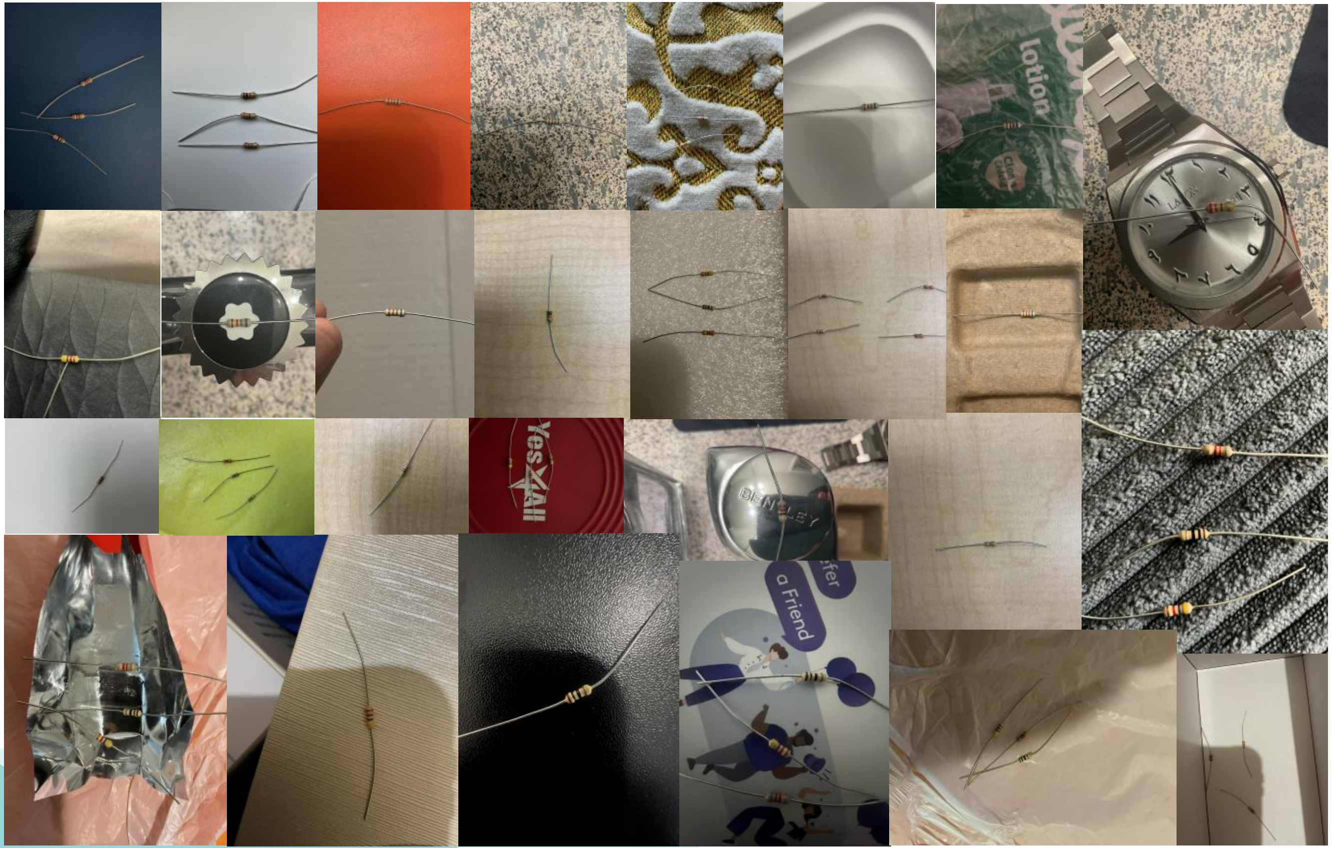
The Cross-Industry Standard Process for Data Mining (CRISP-DM)

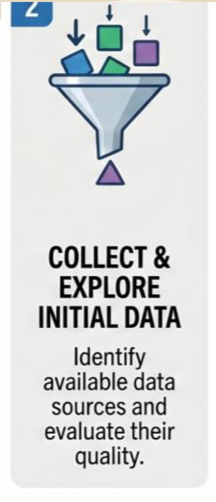


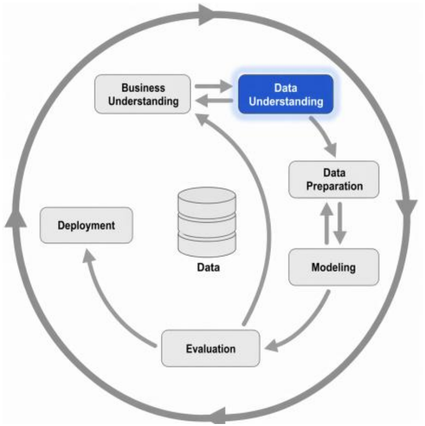
Business Understanding



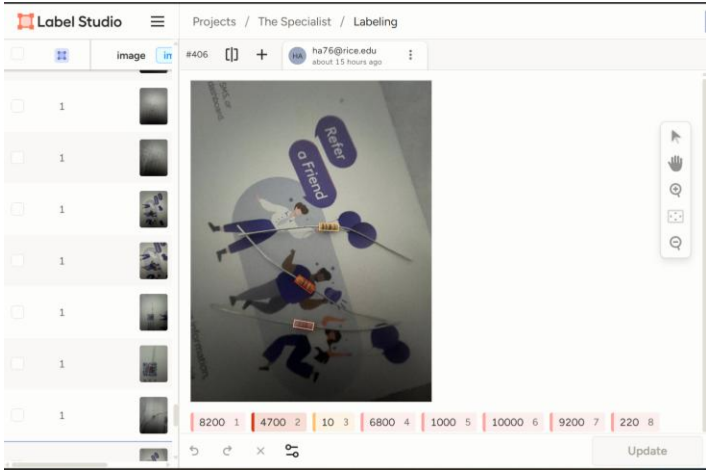
Data Understanding

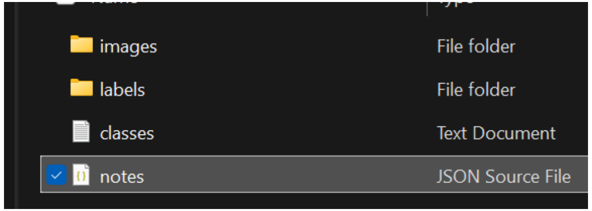


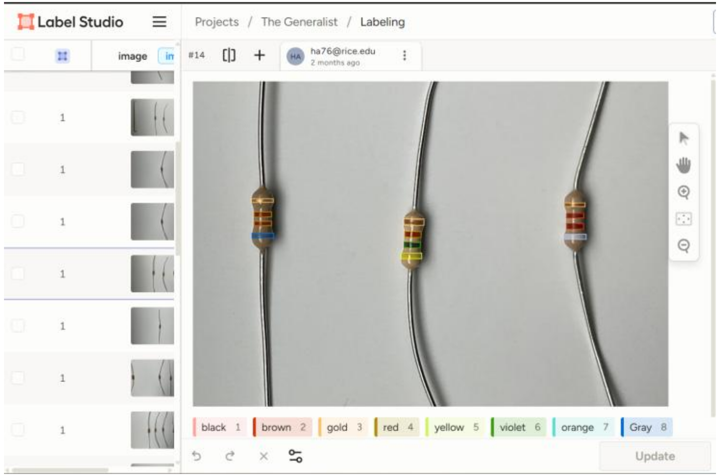




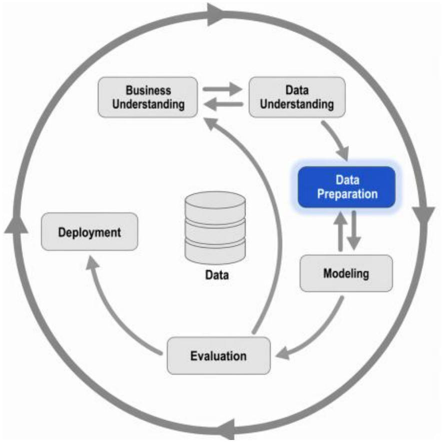
Data Preparation



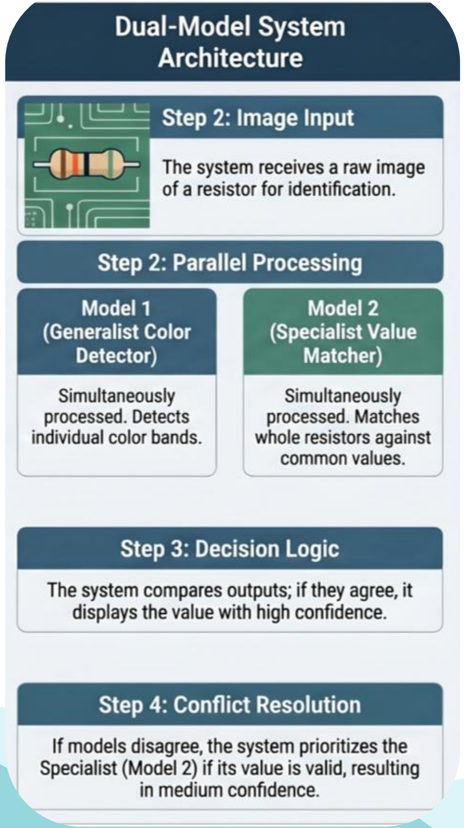




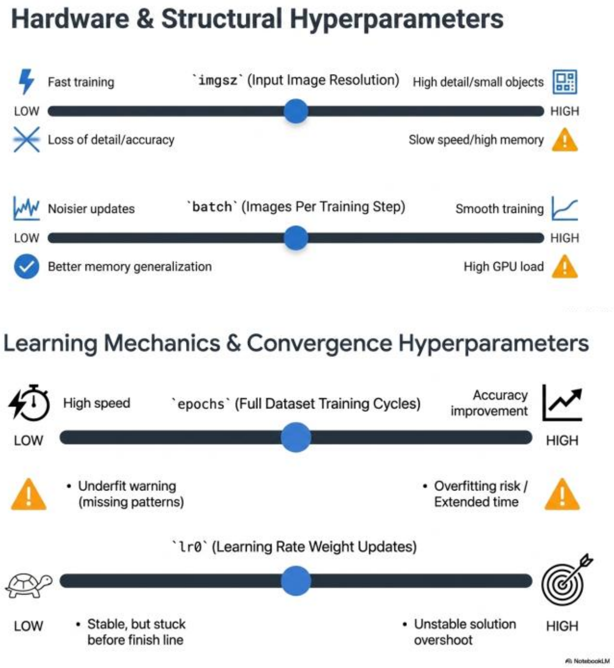
Label Studio



Modeling

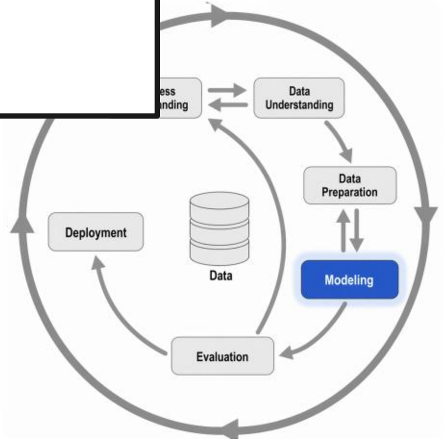


ultralytics
YOLO

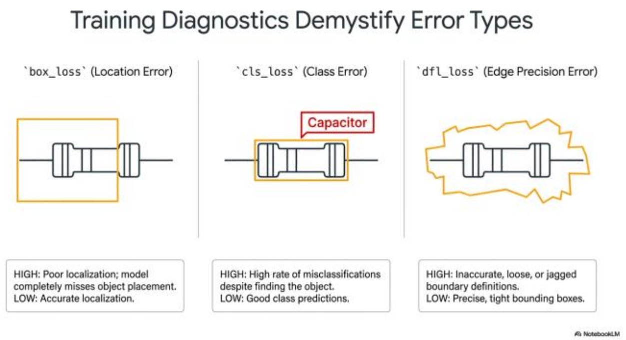


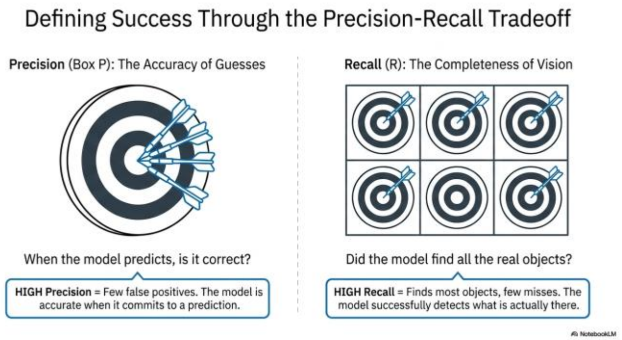
Final tuned hyperparameters will be presented here

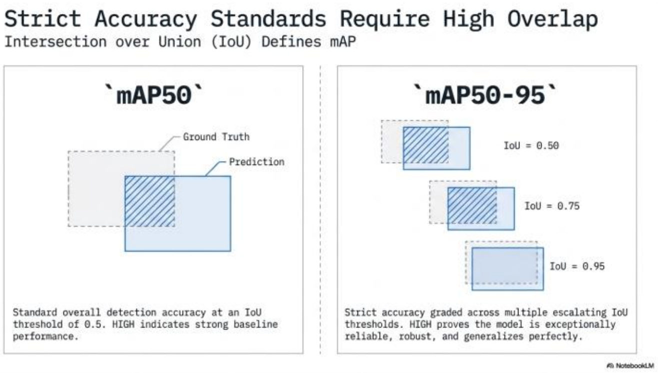
```
epochs=40 \
imgsz=640 \
batch=16 \
lr0=0.005
```



Evaluation



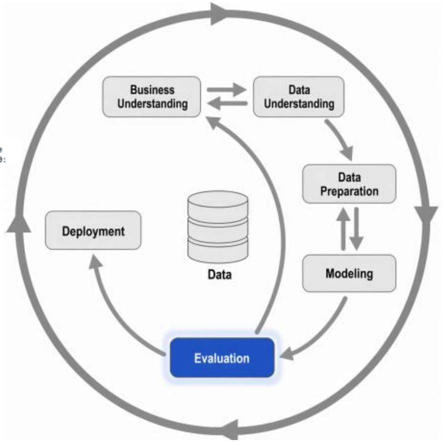


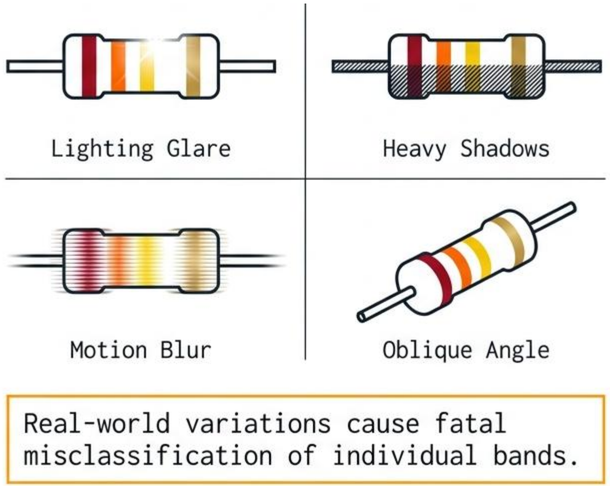


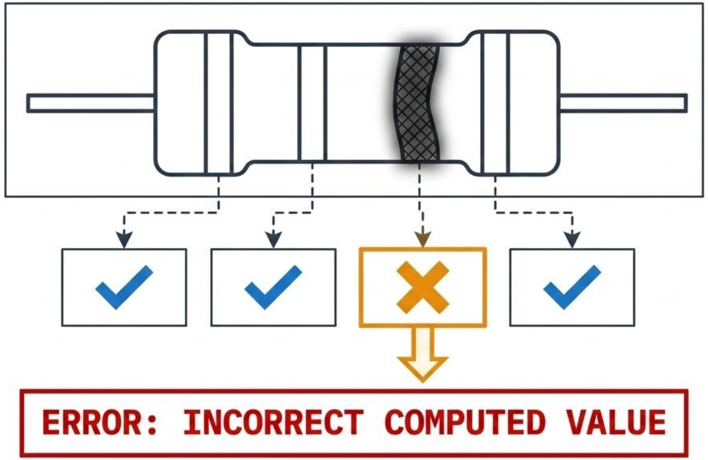
Final evaluation will be presented here

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)
all	59	67	0.972	0.953	0.986	0.821
10	7	7	0.987	0.857	0.925	0.567
1000	7	7	0.973	1	0.995	0.878
10000	10	10	0.981	1	0.995	0.892
20000	12	12	0.984	1	0.995	0.953
220	6	6	0.952	1	0.995	0.919
330	5	5	0.972	1	0.995	0.945
4700	4	4	0.979	1	0.995	0.771
6800	6	6	0.969	1	0.995	0.838
8200	8	8	1	0.669	0.971	0.606
9200	2	2	0.925	1	0.995	0.846

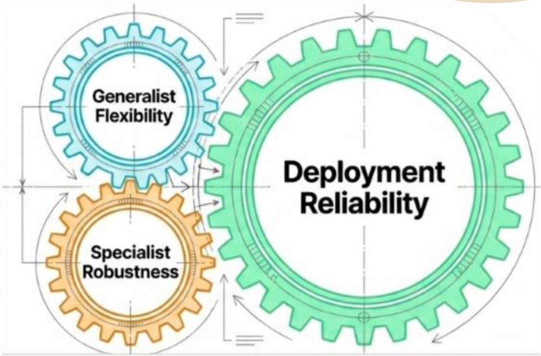
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
40/40	4.87G	0.5788	0.4145	0.8321	11	640:

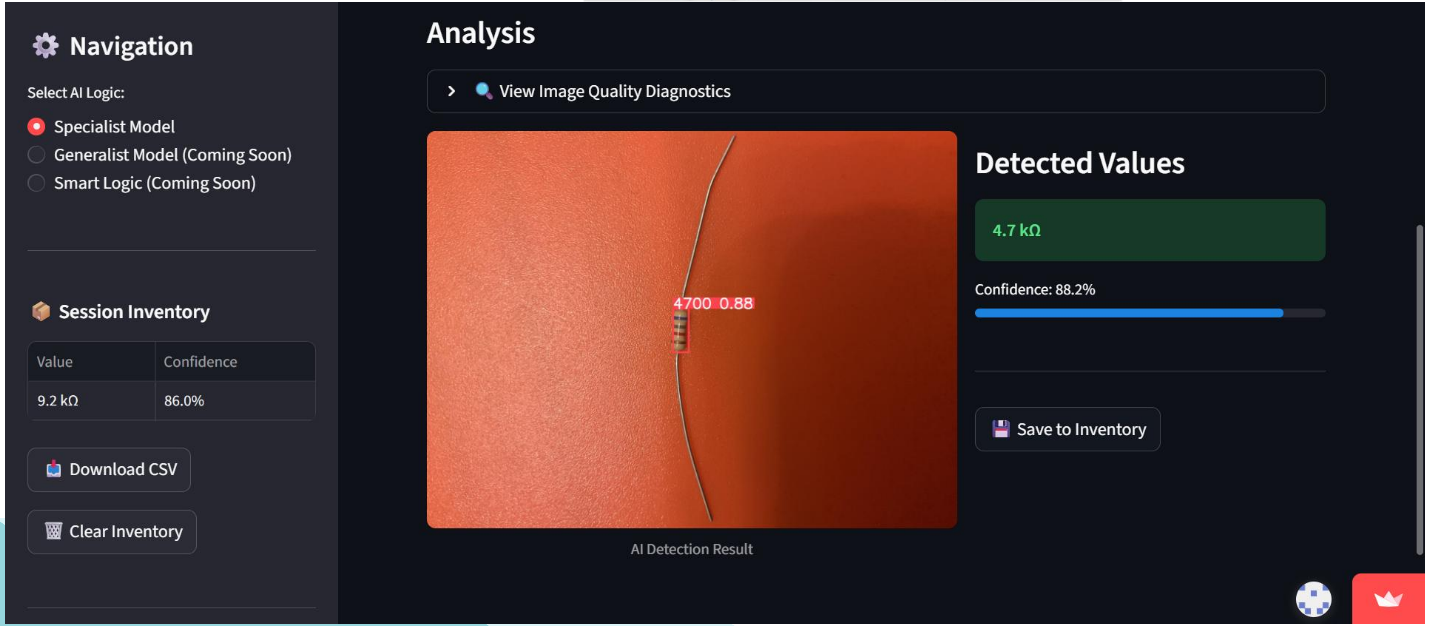


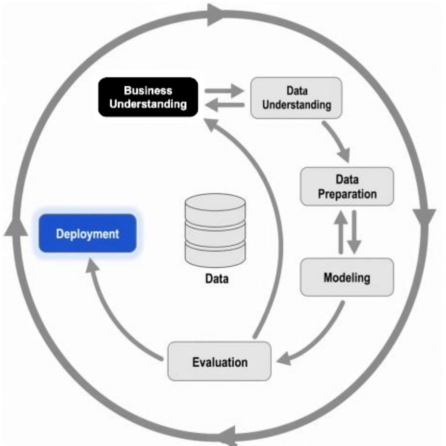

Lighting Glare
Heavy Shadows
Motion Blur
Oblique Angle
Real-world variations cause fatal misclassification of individual bands.


ERROR: INCORRECT COMPUTED VALUE

Deployment?





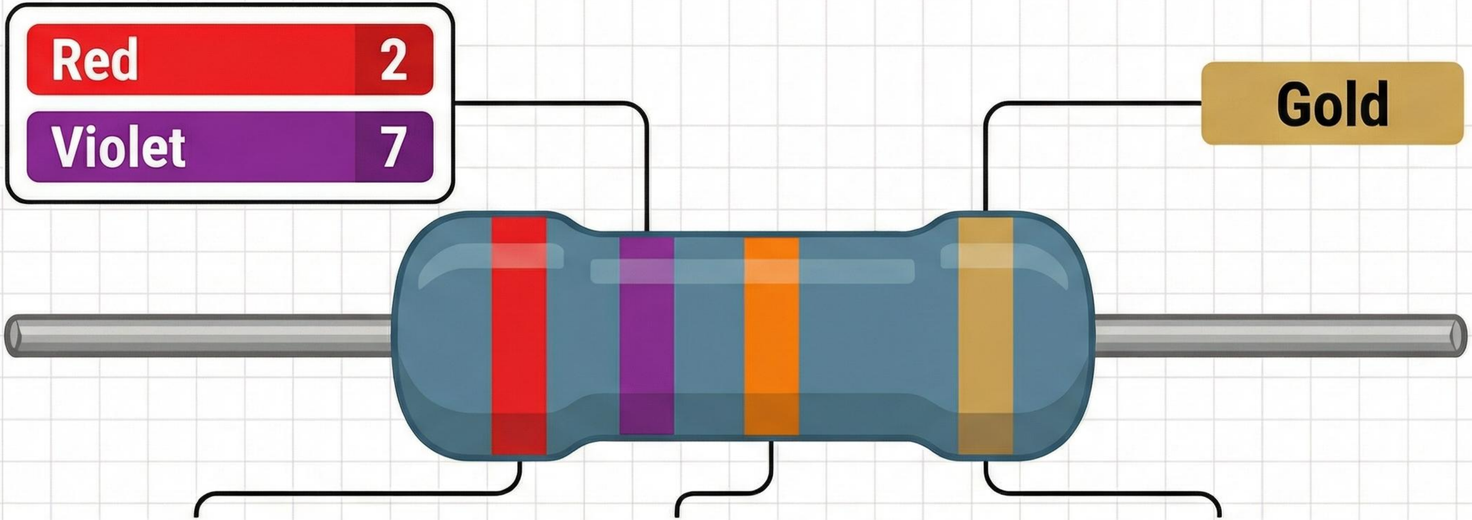


https://resistor-detector-fcfshdi9kgsw59ceu5bmpo.streamlit.app/

Thanks

Do you have any questions?

How to Read Resistor Color Codes



READ FIRST TWO DIGITS

Red (2) & Violet (7) **27**

APPLY THE MULTIPLIER

Orange band = x10^3 (or x1kΩ)

CHECK TOLERANCE BAND

Gold band = ±5% Accuracy

The third band tells you how many zeros to add (e.g., Red = x10^2, or add two zeros).

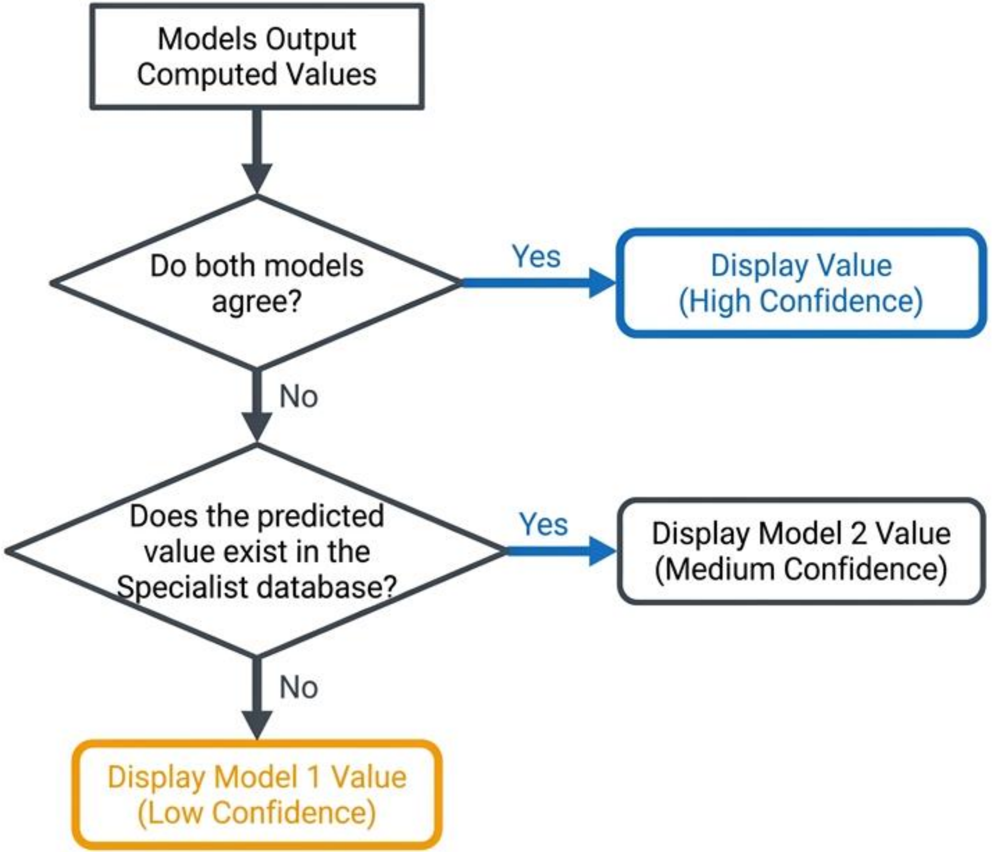
STANDARD COLOR SEQUENCE

- Black (0)
- Brown (1)
- Red (2)
- Orange (3)
- Yellow (4)
- Green (5)
- Blue (6)
- Violet (7)
- Grey (8)
- White (9)

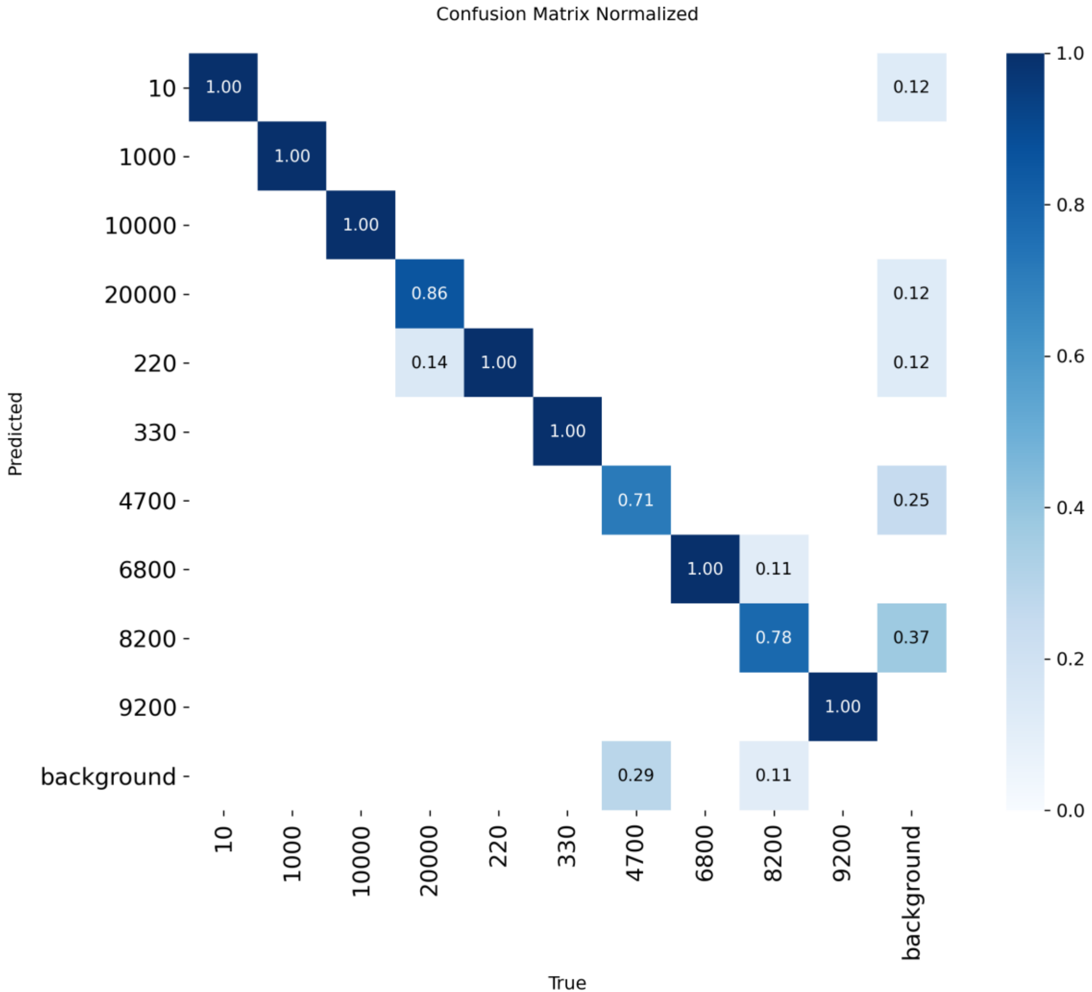
Tolerance

Gold	±5%
Silver	±10%

Smart System Logic Resolves Output Conflicts



The system defaults to high confidence upon consensus, but systematically downgrades trust and defaults to the Specialist when conflicts arise.


Confusion Matrix Normalized
Predicted
True
10
1000
10000
20000
220
330
4700
6800
8200
9200
background
1.00
1.00
1.00
0.86
0.14
1.00
1.00
0.71
0.29
1.00
0.11
0.78
0.11
1.00
0.12
0.12
0.12
0.25
0.37
1.0
0.8
0.6
0.4
0.2
0.0